

Yiyang Huang

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EDUCATION

Northeastern University, Boston, USA

Sep 2024 - Present | Ph.D. in Computer Engineering, advised by [Prof. Yun Raymond Fu](#)

Xidian University, Xi'an, China

Sep 2021 - Jun 2024 | M.S. in Computer Science, advised by [Prof. Xuefeng Liang](#)

Sep 2017 - Jun 2021 | B.Eng. in Intelligence Science and Technology

RESEARCH INTERESTS

Multi-modal LLMs & Agent; Efficiency; Reliability; Hallucination; Video Understanding; Layout Understanding

EXPERIENCE

Northeastern University, Boston, USA

Sep 2024 - Present | SMILE Lab, Research Assistant | Supervisor: [Prof. Yun Raymond Fu](#)

- Submitted 1 **NeurIPS 2026** paper on **Tool-Augmented Agentic Keyframe Selection for Long-form Video Understanding**.
 - Developed SCOUT, an agentic keyframe selection framework that leverages a lightweight VLM (Qwen2.5-VL-3B) as the planning-and-reflection agent and CLIP as the retrieval tool, boosting the long-form VideoQA accuracy of off-the-shelf VideoLLMs (LLaVA-OneVision, Qwen3-VL, InternVL3, and LLaVA-Video) by 5–10% on MLVU, LVB, and VideoMME.
- Published 1 **ACL 2026** survey paper on **Video-LLM Hallucinations**; conducted a comprehensive review and built a curated GitHub repository of related papers and benchmarks. [\[GitHub\]](#)
- Published 1 **ICLR 2026** paper on **Mitigating Hallucinations in MLLMs** via encoder-level inference intervention.
 - Identified encoder-side causes of hallucinations and proposed SHIELD, a training-free module (token reweighting/subtraction + adversarial contrastive decoding), reducing hallucinations by 27% on captioning across MLLM families.
- Published 1 **EMNLP 2025** paper on **Efficiently Scaling Image-pretrained MLLMs to Video Understanding**.
 - Developed D-CoDe, a plug-and-play framework that scales image-pretrained MLLMs to VideoQA via question decomposition and dynamic token/frame compression, improving first-person video understanding accuracy by 29% under LLaVA-NeXT.

Adobe Research, San Jose, USA

May 2026 - Nov 2026 | Vision-Language Lab, Research Intern (Return) | Mentors: [Zhaowen Wang](#), [Shraman Pramanick](#)

- Working on **Agentic Design Layout Evaluation and Refinement**.
 - Identified the lack of holistic layout understanding and document reading-order bias as key challenges in design layout evaluation, and developed a **GRPO-trained layout reflection agent** that provides actionable feedback and refinement suggestions through hierarchical layout understanding and debiased visual reasoning, improving the aesthetic quality of designs generated by Codex and Claude Code.

May 2025 - Nov 2025 | Vision-Language Lab, Research Intern | Mentors: [Zhaowen Wang](#), [Simon Jenni](#), [Jing Shi](#)

- Published 1 **ECCV paper** and filed 1 **patent** on **Compositional Layout Understanding in MLLMs**.
 - Identified semantic drift and structural ambiguity as key challenges in compositional layout reasoning, and developed MASON with metadata-visual grounding and 3D structural perception, improving compositional matching by 26% over Gemini-2.5-Pro and 15% over GPT-5/GPT-o3 under a Qwen2.5-VL backbone.
 - Constructed CoDeLayout, a ~20K-sample benchmark, by developing an automated pipeline for compositional layout mining and QA generation.

PUBLICATIONS

Accepted

- **Yiyang Huang**, Zhaowen Wang, Simon Jenni, Jing Shi, Yitian Zhang, Yizhou Wang, Yun Fu. “Beyond Atomic Layouts: Compositional Design Understanding with Vision-Language Models.” ECCV 2026. [\[Project Page\]](#)
- **Yiyang Huang**, Yitian Zhang, Yizhou Wang, Mingyuan Zhang, Liang Shi, Huimin Zeng, Yun Fu. “Distorted or Fabricated? A Survey on Hallucination in Video LLMs.” ACL 2026 (Findings). [\[Paper\]](#) [\[Repo\]](#)
- **Yiyang Huang**, Liang Shi, Yitian Zhang, Yi Xu, Yun Fu. “SHIELD: Suppressing Hallucinations in LVLM Encoders via Bias and Vulnerability Defense.” ICLR 2026. [\[Paper\]](#) [\[Code\]](#)
- **Yiyang Huang**, Yizhou Wang, Yun Fu. “D-CoDe: Scaling Image-Pretrained VLMs to Video via Dynamic Compression and Question Decomposition.” EMNLP 2025 (Main). [\[Paper\]](#) [\[Code\]](#)
- **Yiyang Huang**, Xuefeng Liang, Chaowei Fang. “CALLip: Lipreading using Contrastive and Attribute Learning.” ACM MM 2021. [\[Paper\]](#)

- Liang Shi, **Yiyang Huang**, Yun Fu. “Capturing Individual Differences of Facial Expression for Authentic Expression Generation.” IEEE FG 2026.
- Shuai Zou, Xuefeng Liang, **Yiyang Huang**. “LipReading for Low-resource Languages by Language Dynamic LoRA.” ICASSP 2025. [\[Paper\]](#)

Preprint / Under Review

- **Yiyang Huang**, Yitian Zhang, Yizhou Wang, Jianglin Lu, Qihua Dong, Hailing Wang, Huimin Zeng, Yun Fu. “SCOUT: Selecting Critical Observations via Tiny VLM Planning and Tool-Augmented Retrieval.” under-review.
- Mingyuan Zhang, Yue Bai, Yifan Wang, **Yiyang Huang**, Yun Fu. “Rethinking Fine-Tuning: Unlocking Hidden Capabilities in Vision-Language Models.” preprint. [\[Paper\]](#)
- Hailing Wang, Jianglin Lu, Xu Ma, **Yiyang Huang**, Yun Fu. “StreamVLM: Adapting Offline Vision-Language Models for Streaming Understanding.” under-review.
- Huimin Zeng, Yue Bai, Qihua Dong, Hailing Wang, **Yiyang Huang**, Yun Fu. “Narrative3D: Asset-to-Storyworld Generation.” under-review.

ACADEMIC SERVICE

Conference Reviewing: NeurIPS; ACL; EMNLP; IEEE Face & Gesture (FG)

Journal Reviewing: ACM TKDD

TEACHING

Teaching Assistant

DS 5110: Essentials of Data Science

Fall 2025

DS 5020: Fundamentals of Linear Algebra and Probability

Spring 2026

SKILLS

Languages: Python, $\LaTeX$

Frameworks/Tools: PyTorch, HF Transformers, HF Datasets, ffmpeg, PyAV, decord, Git, Slurm

Training/Finetuning: Accelerate, PEFT, LoRA/QLoRA, bitsandbytes

Inference: vLLM, FlashAttention

Evaluation: Imms-eval, POPE, CHAIR, MME

AWARDS

National Scholarship, China

2021

Outstanding Student, Xidian University

2022

1st Prize, Undergraduate Computer Design Competition (National Level), China

2021

2nd Prize, RoboMaster National Robotics Competition, China

2019

3rd Prize, ICRA AI Challenge

2019